

Courses

9–11 minutes

The Computer Science curriculum is traditional, at least in the early courses. The first two courses in the major are taught in Java, which is an excellent language for introducing object-oriented concepts.

The following list of courses represents current or recent course offerings. See the course [catalog](#) for updated information.

- 54-144 Explorations in Computing

An introduction to the discipline of computer science with an emphasis on applications in and connections to the liberal arts and everyday life. Programming topics include: basic programming constructs, basic data structures, algorithmic computation, selection, iteration, abstraction and reasoning about computer programs. This is an introductory course intended for humanities, social science and fine arts majors. May not be used for the Computer Science major or minor. Cannot be taken concurrently with or after successful completion of Computer Science 54-184, 54-284, or 54-454 without departmental approval. Contributes to Data Analytics and Data Science. (NS)

- 54-184 Computer Science I

This first course in computer programming in an object-oriented style is primarily intended for students pursuing a major or minor in computer science, mathematics or other disciplines in the natural sciences. Topics include: primitive types and operations,

assignment, conditional execution, iteration, arrays, classes, methods, recursion, encapsulation, type extension, inheritance and reasoning about programs in Java. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Contributes to Data Analytics and Data Science. (Fall, Spring) (NS)

- 54-284 Computer Science II

This course is a continuation of 54-184 Computer Science I, with an emphasis on abstract data objects such as lists, stacks, queues, trees and graphs, and test-first programming using the automated testing paradigm JUnit. Topics include: algorithms for searching, sorting, traversing, inserting and deleting, and reasoning about the run-time complexity of these algorithms. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Contributes to Design Thinking.

Prerequisite: Computer Science 54-184 or permission of instructor. (Spring) (NS)

- 54-291 Puzzling Programs

Given a precise program specification and a few input/output examples, students will gain experience quickly devising a solution that passes a suite of unseen (and generally more complicated) input/output examples. Teamwork, troubleshooting code based on very limited feedback from automated systems, effectively reading problem descriptions, and discerning relevant details will be practiced. This course may be repeated for credit, but may not be counted towards the major or minor. Must be taken Pass/D/F.

Prerequisite: Computer Science 54-184. (Fall)

- 54-384 Discrete Mathematics

A study of concepts in the modeling of discrete phenomena. Topics include: logic, structure of proof, set theory, combinatorics, graphs, induction and recurrence relations. Also Mathematics 52-384.

Prerequisites: Mathematics 52-164, Computer Science 54-184, and either Mathematics 52-264 or Computer Science 54-284, or permission of instructor. (Fall) (NS)

- 54-394 Computer Organization

A study of computer architecture, internal representation of data, assembly language programming, subroutines and parameter passing, design of machine language instruction sets, bus structure, digital logic, and support for parallelism. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Prerequisite: Computer Science 54-284 or permission of instructor. (Fall) (NS)

- 54-414 Operations Research

Formulation and solution of problems with management, economics, engineering and science applications using modeling, optimization techniques, and simulation. Topics include: linear and integer programming, simplex method, duality, sensitivity analysis, and problem solving using optimization software. A significant group project resulting in a written report and presentation is required. Also Mathematics 52-414 and Business 30-414.

Contributes to Data Analytics and Data Science. Prerequisites: Mathematics 52-164 (Modern Calculus I), and either Business 30-474 (Finance), Mathematics 52-674 (Linear Algebra), some Computer Science course at the 300 level or above, or permission of the instructor.

- 54-424 Artificial Intelligence

A study of the use of computer algorithms to emulate the cognitive and problem solving capabilities of humans. Topics include: search, multi-agent search, optimization, reinforcement learning, evolutionary computation, statistical methods, (deep) neural networks, and generative artificial intelligence. Contributes to Neuroscience. Prerequisites: Mathematics 52-164 and Computer

Science 54-284, or permission of instructor. (NS)

- 54-454 Algorithms

An introduction to the design and analysis of algorithms and their implementation in C++. Topics include: greedy, divide-and-conquer, dynamic programming, graph, and randomized algorithms; asymptotic analysis and computational tractability (e.g. NP-completeness); practical data structures (stacks, trees, queues, graphs, hash tables, heaps). Prerequisites: Mathematics 52-164 and Computer Science 54-284, or permission of instructor.

Contributes to Design Thinking. (Spring) (NS)

- 54-474 Programming Languages

A study of the history, principles, and practice in the design and implementation of imperative, functional, logic and object-oriented programming languages. Topics include: syntax, context-free grammars, scope, data types, control structures, sub-programs, exception handling, and concurrency. Prerequisite: Computer Science 54-394 or permission of instructor. (Fall) (NS)

- 54-514 Database Management

An introduction to the theory and practice of database management systems. Topics include: database terminology, the entity-relationship model, the relational model, normalization, querying databases using SQL, NoSQL models, and other database technologies. The course includes a laboratory component designed to explore applications and to enhance conceptualization. Contributes to Data Analytics and Data Science. Prerequisite: Computer Science 54-284 or permission of instructor. (NS)

- 54-524 Introduction to Numerical Analysis

See Mathematics 52-524. (NS)

- 54-534 Functional Programming

An introduction to functional programming. Topics include functions, lists, types, induction and recursion, pattern matching, infinite lists and trees. A functional programming language such as Haskell, Lisp or ML will be used in the course. There will be a large number of programming projects. Prerequisite: Computer Science 54-284 or permission of instructor. (NS)

- 54-644 Computer Systems

An introduction to operating systems and computer networks. Topics include process control, scheduling, threads, concurrency, memory management, virtual memory, network protocol layers, packets, routing and network security. Prerequisite: Computer Science 54-394. (Spring) (NS)

- 54-684 Theory of Computation

An introduction to the theory of information and computation as physical phenomena, covering standard formalizations of computational concepts and proofs of their implications. Includes a programming component emphasizing efficient design and implementation of course concepts. Topics include: formal languages, finite automata, regular expressions, context-free grammars, pushdown automata, Turing machines, decidability, and reducibility. Prerequisites: Mathematics 52-164 and Computer Science 54-284, or permission of instructor. (NS)

- 54-844 Seminar in Special Topics

A limited enrollment seminar not generally covered in other courses. This course may be repeated for credit as topics vary. (NS)

- 54-894 Senior Seminar in Software Engineering

Introduction to techniques and theories for the development of large software systems. This course will fulfill the capstone requirement in Computer Science. Topics include: software design and quality,

ethics, professional issues, the study of current software engineering trends, theory and practice. Each student will contribute substantially to a major semester project as part of a software development team, give at least one major individual presentation, and have significant class participation. Each student will take the Major Field Test. Prerequisite: Five courses in the Computer Science major at the 300 level or above, and permission of instructor. Must have completed one of Computer Science 54-454, 54-514, or 54-474. (Spring) (NS) (WA)